Machine Learning-Powered Tools to Elucidate Species-Specific Promoter Design Rules Across the Proteobacteria

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Abstract

Despite decades of research into bacterial promoters, accurately predicting promoter activity from DNA sequence remains a challenge, even in model organisms. In the absence of systematic studies, there is a general assumption that the same promoter sequence will have similar activity across closely related species. However, research has shown that when the same promoters are transferred among Proteobacteria within the same taxonomic class, their rank order of expression is often inconsistent. Uncovering a phylogenetic correlation in promoter activity, where closely related species exhibit similar promoter activity levels, could guide the discovery of effective promoters in non-model species.

Introduction

Promoters are essential regulators of gene expression and have immense utility as genetic tools. Although much of our understand of promoters comes from model species, species important to agriculture and human health are increasingly employed as cellular chassis, enabling researchers to exploit their metabolic diversity. Current methods for identifying effective promoters in non-model species are inefficient and overlook species-specific promoter activity determinants. Uncovering phylogenetic correlations in promoter activity, where closely related species exhibit similar promoter activity levels, could streamline this process, enabling a more targeted approach. Artificial intelligence can elucidate the complex biophysicochemical relationships contributing to promoter activity and provide insight into phylogenetic correlations. Here, we screen a promoter library in 15 species of Proteobacteria and use that expression data to train machine-learning models specific to Alpha-, Beta- and Gammaproteobacteria, uncovering significant differences in feature importance profiles.

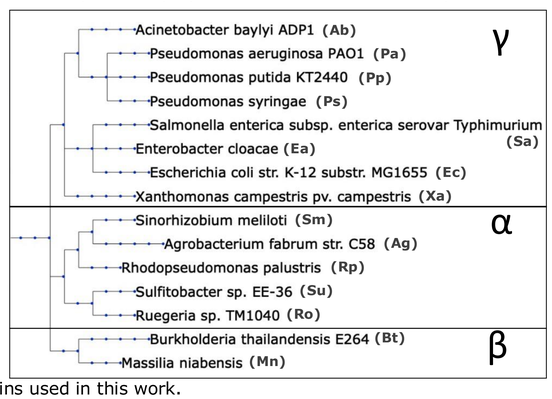


Figure 1. The 15 bacterial strains used in this work.

Methods

Here, we employed high-throughput screening to measure expression from a promoter library in 15 Proteobacterial species in the Alpha-, Beta-, and Gammaproteobacteria to generate a comprehensive expression dataset. We first generated feature vectors that encoded biochemical, biophysical and compositional properties of the sequences. We leveraged these encodings to train machine learning classifiers for each taxonomic class and achieved an AUC score exceeding 0.75 in all three models. To uncover phylogenetic relationships, we tested the performance of models trained on data from a single taxonomic class in classifying promoter activity from species of that same class compared to species from other classes. Finally, we applied explainable AI for model interpretation to compare feature importance profiles across the three models to elucidate species-specific promoter design rules. This work will enable a more targeted approach to identifying promoters with predictable activity across species, streamlining genetic circuit design for a range of genetic engineering applications.

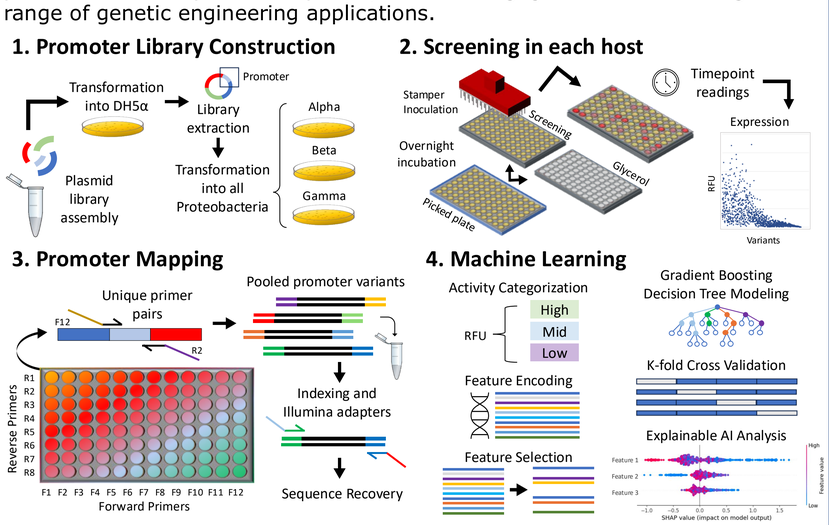


Figure 2. Visual abstract outlining workflow. Plasmid construction followed our toolbox protocol[1] and constitutive promoter library was a generous gift from the Salis Lab[2].

Results

Inconsistent Promoter Activity Across Proteobacterial Species

To systematically evaluate the consistency of promoter activity levels across species, we selected a range of characterized promoters from each of four hosts, including 12 promoters from *E. coli*, *B. thailandensis*, *P. aeruginosa*, and *S.* sp. EE-36. We then screened the promoters selected from the three other species in the remaining host, ensuring that all 47 promoters were characterized in all four hosts. We plotted the rank order expression level changes for each promoter set on slopegraphs. We used Spearman correlation to assess the relationship between rank order of promoter activity levels across the species.

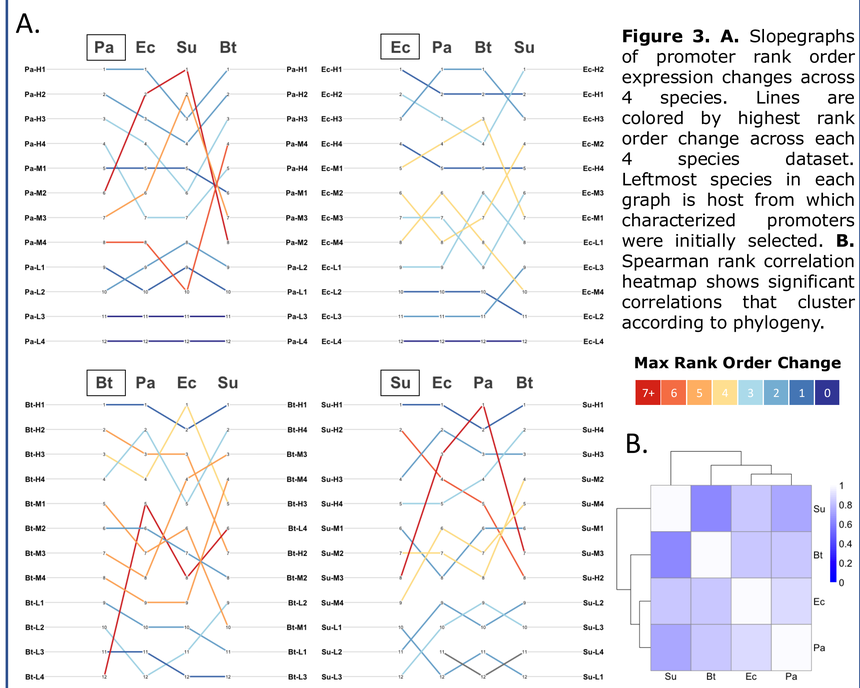


Figure 3. A. Slopegraphs of promoter rank order expression changes across 4 species. Lines are colored by highest rank order change across each 4 species dataset. Leftmost species in each graph is host from which characterized promoters were initially selected. **B.** Spearman rank correlation heatmap shows significant correlations that cluster according to phylogeny.

Performance of XGBoost Classifier Models

To model promoter activity, we first encoded promoter sequences into over 40 feature descriptor types[3] and used mutual information scores and SHAP (SHapley Additive exPlanations) values for feature selection. Between 20 and 22 features were used to train each final model. For each species, promoter data was split into high, mid, and low activity categories, in a 25/50/25 ratio, respectively. For each species group, we trained models on promoter data from each activity category for better resolution of the features important for promoters of varying strengths. All models have an AUC above 0.93 and low activity models have an F1-score of 0.72 or higher, though high and mid activity models had F1-scores above 0.82.

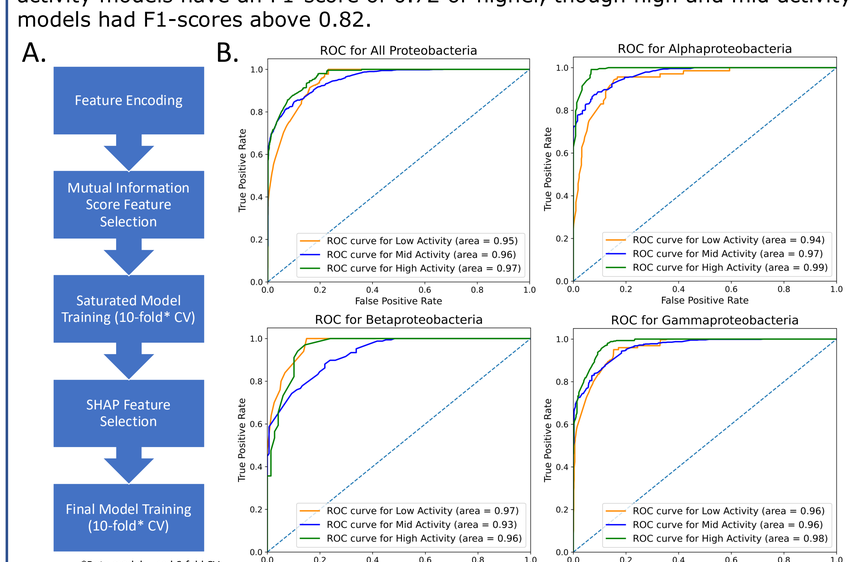
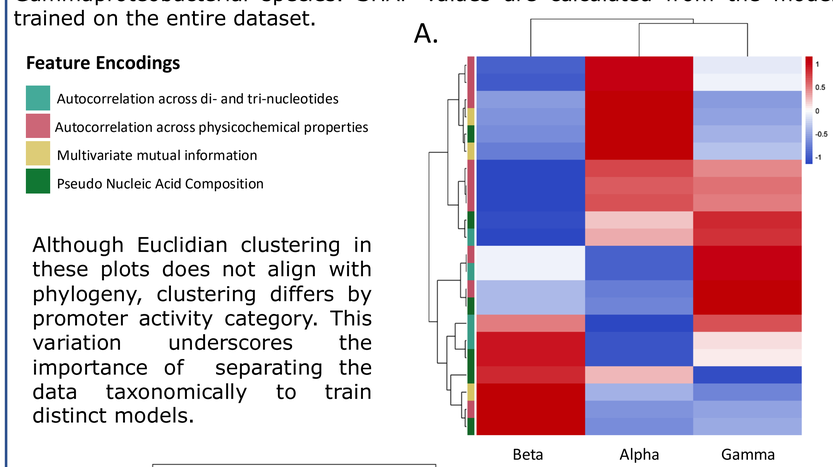


Figure 4. A. Machine learning workflow. Features were encoded using the iLearn platform[3] and two feature selection steps were used prior to final model training. **B.** ROC/AUC plots for 12 XGBoost models after 10-fold cross validation (CV). Models were trained on low, mid, and high activity promoters for each species group. Species groups include activity data from all Proteobacteria and species from the Alpha-, Beta-, and Gammaproteobacterial classes.

Feature Importance Across Phylogenetic Class and Species

To show variation in feature importance profiles across classes and species, we plotted summed absolute SHAP values of the union of features from the mid activity promoter models across all Proteobacterial classes and Alpha- and Gammaproteobacterial species. SHAP values are calculated from the model trained on the entire dataset.



Although Euclidian clustering in these plots does not align with phylogeny, clustering differs by promoter activity category. This variation underscores the importance of separating the data taxonomically to train distinct models.

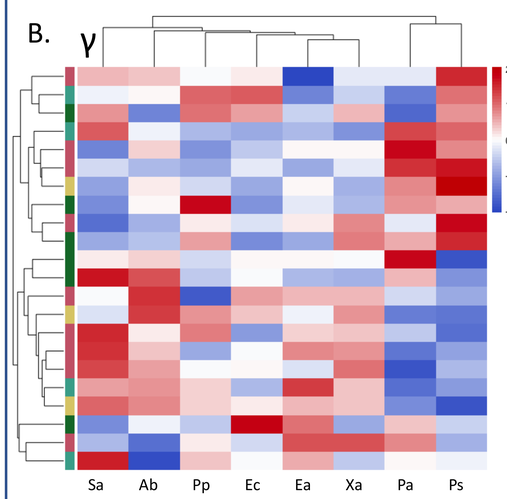
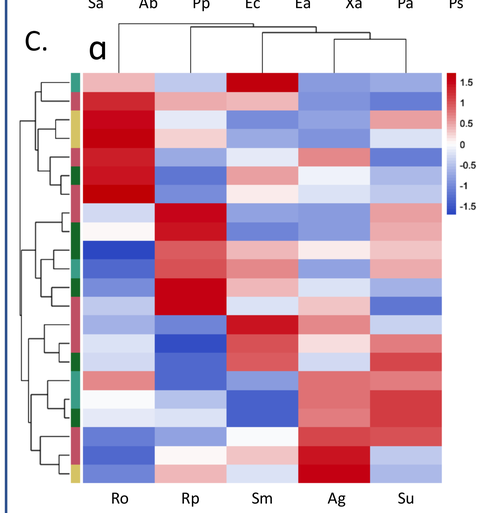


Figure 5. Heatmaps are generated from the summed absolute SHAP values generated from the final model and Z-score normalized across columns and rows. SHAP data are from the mid-activity promoter category. (**A.**) SHAP values across the Alpha-, Beta-, and Gammaproteobacteria subsets, (**B.**) across each species from the Gammaproteobacterial class, and (**C.**) the Alphaproteobacterial class. Pairwise comparisons using Wilcoxon signed-rank tests on pre-normalized data across columns showed that all comparisons were statistically significant in **A.**, 23/28 comparisons were significant in **B.** and 9/10 comparisons were significant in **C.**



Many features across models, including autocorrelation and cross-covariance, highlight that the interdependencies among sequence positions, rather than isolated positions, are important determinants of promoter function. Pseudo nucleic acid composition features combine frequency data with global sequence-order information, suggesting that both content and context are key. The top two features across the 3 classes (**A.**) are those encoding the autocorrelation of the roll, or the twisting around the longitudinal axis of DNA, and minor groove depth.

Conclusions and Future Directions

Here, we have identified important and unique promoter features across three Proteobacterial classes and developed a tool for modulating gene expression. The promoters from our dataset can also be reutilized directly. Our future steps involve hyperparameter tuning, model testing on independent validation sets, and cross group model testing to demonstrate model specificity.

References

1. Schuster, Layla A, and Christopher R Reisch. "A Plasmid Toolbox for Controlled Gene Expression across the Proteobacteria." *Nucleic Acids Research* 49, no. 12 (2021): 7189–7202.
2. Hossain, Ayaan, Eriberto Lopez, Sean M. Halper, Daniel P. Cetnar, Alexander C. Reis, Devin Strickland, Eric Klavins, and Howard M. Salis. "Automated Design of Thousands of Nonrepetitive Parts for Engineering Stable Genetic Systems." *Nature Biotechnology* 38, no. 12 (2020): 1466–75.
3. Chen, Zhen, Pei Zhao, Chen Li, Fuyi Li, Dongxu Xiang, Yong-Zi Chen, Tatsuya Akutsu, et al. "iLearnPlus: A Comprehensive and Automated Machine-Learning Platform for Nucleic Acid and Protein Sequence Analysis, Prediction and Visualization." *Nucleic Acids Research* 49, no. 10 (June 4, 2021): e60.

Acknowledgements

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